CS411

Project Report

Team020: bigOenergy

- **Please list out changes in the directions of your project if the final project is different from your original proposal (based on your stage 1 proposal submission).**

  At first, the project was going to be an NCAA Division 1 basketball analytics dashboard. But we changed the direction to focus on the NBA because we thought that there was more data and information on the NBA for this project. The final project became an NBA analytics web application connected to a MySQL database, instead of the broader NCAA idea from the beginning.

- **Discuss what you think your application achieved or failed to achieve regarding its usefulness.**

  I think our application achieves the goal of making all NBA data easier to access in one place. The application and the database help organize player, team, and game data. I think this method is better than looking at CSV files or Excel sheets and trying to figure out information yourself. It is quick, accessible, and efficient. One thing we failed to achieve is to have more advanced functionalities. We initially thought of incorporating more advanced functions, but as the semester progressed, we realized we should stick to simpler methods.

- **Discuss if you changed the schema or source of the data for your application.**

  We changed both the source and the schema direction of the project. Since the project changed from NCAA basketball to NBA basketball, the data source and overall database design also changed. Instead of building around NCAA data, we built our schema around NBA-related entities. The schema became more focused on NBA player, team, and game analytics and was implemented in MySQL on GCP. For example, some of our main tables included Game, Players, and Teams, and we had relationship tables such as PlayerGameStats and TeamGameStats to capture interactions with our entities. We also tried to normalize and reduce any redundancy.

- **Discuss what you changed in your ER diagram and/or your table implementations. What are some differences between the original design and the final design? Why? What do you think is a more suitable design?**

  We changed parts of the ER diagram and table design after we got feedback on Stage 2. The earlier version had some mistakes in the diagram. The final version kept the main NBA entities, but we cleaned the associative tables like PlayerGameStats, TeamGameStats, and GameOfficial. We also made the schema easier to support our queries for later in the project.

- **Discuss what functionalities you added or removed. Why?**

  The final project focused more on retrieval, keyword search, analytics queries, and connecting the frontend, backend, and MySQL database together. In the beginning, we had some complicated plans like a predictive tool, more customizable comparisons, and broader smart suggestions. We ended up removing them to focus on features that were realistic to complete and demo correctly.

- **Explain how you think your advanced database programs complement your application.**

  I think the advanced database programs help the application because they make the project more than just a basic website connected to tables. The advanced SQL work gives the application more useful basketball analysis. These queries are more meaningful because they help users filter data, unlike other websites that just give you raw data or information, and provide no type of way to search for what you are looking for.

- **Each team member should describe one technical challenge that the team encountered. This should be sufficiently detailed such that another future team could use this as helpful advice if they were to start a similar project or where to maintain your project.**

  Eldiar: Deploying the built React + Flask app on GCP Cloud Run took extremely long because there were no proper tutorials online. I would say at least 3-4 hours.

  Jovani: Fully switching the frontend from basic JavaScript React to TypeScript with libraries like reactstrap and @tanstack/react-query to enhance the website's design.

  Alexis: Distributing IAM roles and permissions to team members so they could access the database, and spending a lot of time troubleshooting why Eldiar couldn't deploy to Cloud Run even after being granted a bunch of different roles

  Diego: Getting indexing to work on GCP SQL when only the admin user had permissions, and understanding why EXPLAIN ANALYZE showed no cost difference across index configurations (it was due to the optimizer's choices).

- **Are there other things that changed when comparing the final application with the original proposal?**

  The original proposal had a bigger focus on things like smart suggestions, prediction features, and customizable comparisons. The final project became more focused on the core database application, which included connecting the frontend, backend, and MySQL database, supporting CRUD and keyword search, and building advanced SQL features that could be used from the application.

- **Describe future work that you think, other than the interface, that the application can improve on**

  We can expand the advanced database features so they are more deeply connected to the application. Also have stronger stored procedures, triggers, and transactions that support more realistic user actions. Then we can add more search options so users can do more detailed player, team, and game comparisons. Fixing the backend logic and allowing for larger amounts of data. I think we can also try to implement the advanced functions we wanted in the beginning, such as smart suggestions, prediction features, and more customizable comparisons. This would make our application even more useful and easier to use in the future.

- **Describe the final division of labor and how well you managed teamwork.**

  We divided the work among team members based on the key components of the project, including frontend, backend, SQL queries/database components, and documentation/deliverables. Jovani worked on the backend, Eldiar focused on the frontend, Alexis handled SQL queries and documentation/deliverables, and Diego worked on SQL queries and documentation/deliverables. Everyone worked on database components. While everyone had their main roles, we helped each other as much as we could. Overall, our team collaborated very effectively. We were consistent with our communication by having conversations in our group chat, as well as holding meetings every week, either in person or virtual, to discuss ideas and opinions about our project. We used each other's strengths and tried to support each other as best we could. This allowed us to stay organized and efficient. This also shows that we had good teamwork and maintained good team chemistry throughout the project.